Report 30/40+12/20 code+28/30 presentation=80/100 project score

The requirements are

Report:

1) Your report must be typeset in LaTeX, use the RevTeX style manuals used by Physical Review Journals.

5 nice job

2) Figures should have clear labels and captions. All figures should be referred to in the text and tell part of the narrative.

~~Captio~~ 3 good job

but more discussion of lines / emission / absorption needed in left x-ray

3) Your paper should have a clear abstract, no more than a paragraph, that highlights what you have done.

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4) You should explain the context and motivation of the physics problem you are solving.

~~4~~ 7 well done

but more discussion SFR, composition, line physics needed

5) You should write-down all equations solved by your code in the report.

2: good overall structure but expect expectation values / linear algebra not entirely correct orthogonality / orthonormality / random PCA not fully or clearly explained.

6) You should explain the computational methods used to solve these equations – if they are methods that were discussed in class, say so! You are welcome to use libraries (e.g. scipy) instead of writing things from scratch, but be sure to acknowledge all libraries used.

3 (See above)

7) Have a discussion or figure showing that your code can reproduce a classic test case or known solution If you failed to reproduce published results or classic test cases, please hypothesize why.

8) If your code produces some numerical result (e.g. the binding energy of something), have the result stated clearly in a table. If you noticed an interesting trend, have clear figures showing that trend.

9) You should have a clear reference list, typeset using BibTeX, citing all methods/outside software packages used.

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10) You must read and draw on 2 primary physics literature references (if you are doing a topic I suggested, these must go beyond the suggested reading), and as many pedagogical/textbook references as you like.

3 good, but dersor discussion needed.

11) If your code takes more than a few minutes to run, explain the computational challenges of your project in the report.

N/A

12) Make sure to acknowledge your team and collaborators (and anyone you sought advice from) in your report.

Code:

1) Does the code run?

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2) Is the code well structured?

3— Straightforward implementation - but more tasks should have been taken on (4-5 assignment worth of work in project description). More extensive code effort needed

3) Is the code well documented?

3 — good documentation of your own contributions but sparse documentation of substantial reliance on publicly available libraries

4) Is the code self-sufficient?

1— sklearn and astroML needed but no expalnation

You should prepare a keynote/Powerpoint style presentation on your work. This presentation should

1) Lay out the motivation for the physics problem to be solved.
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2) Lay out the equations to be solved.
   4 confusion on random PCA
3) Have helpful schematics to explain the problem.
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4) Name the computational methods used or to be used – if they were covered in class, review them. If not, briefly explain them.
   See #2
5) Explain what code you have written and what it is intended to do.
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6) Explain if your code is working. If yes, show some preliminary results (preferably as a graph or animation) and discuss if they make sense to you. Why or why not? What do you plan to try in this project? What are the next steps? If not, what are you doing to troubleshoot your code and what are the major sticking points?
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SDSS overview with a huge survey summar. 930,000 gals, 4000 spectra with 1000 bins
Nice job highlighting data set *Jake VanderPlas (UW)*

Querry straight from SDSS. **Good overview of spectra in general**.

Blooper, ionize matter, not big deal, but mention in written feedback.

Neat idea – curse of dimenionality – hard to predict a simple model for a large data set – reduce the dimensionality to only the most important variable.

Discussion of eigen-decomposition. Eigenvalues are a rating system – **good discussion of linear algebra and how it connects to stuff.**

Sklearn – offers a PCA class that performs the bulk computation. RanomdizedPCA – minimization of weight loss function. A lot of the fluctuations are captured by the first principle components. Higher PCs behavior flattens out, aside . Reconstruction.

First few pcas PCA explain 93% of data variation

Use PCA to fill in missing data.

Clear goals – reconstruction function.

Look at how eigenmades .

How well does PCA work?